

Febrile seizures in the developing brain result in persistent modification of neuronal excitability in limbic circuits

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Febrile (fever-induced) seizures affect 3-5% of infants and young children. Despite the high incidence of febrile seizures, their contribution to the development of epilepsy later in life has remained controversial. Combining a new rat model of complex febrile seizures and patch clamp techniques, we determined that hyperthermia-induced seizures in the immature rat cause a selective presynaptic increase in inhibitory synaptic transmission in the hippocampus that lasts into adulthood. The long-lasting nature of these potent alterations in synaptic communication after febrile seizures does not support the prevalent view of the 'benign' nature of early-life febrile convulsions.